

Date:

EXPERIMENT NO. 7**AIM: Traffic Data Analysis using Wireshark.**

Tools required:

- Desktop computer
- Wireshark

Submission: Written file submission (starting from page no. 4) is expected.

ReferencesWireshark Installation : [Wireshark Tutorial for Beginners - Installation](#)Wireshark Walkthrough : <https://www.youtube.com/watch?v=lb1Dw0elw0Q>**Wireshark Videos:**

- Edit Windows
- Canvas Window
- Capturing Data
- Interpreting data in Wireshark
- Ethernet Frame
- IP Frame
- Throughput
- Flow diagram

Understanding Require

- IP Address
- URL
- Finding IP address from URL
- Finding location from IP Address

Wireshark for Windows**Install wireshark from below link**<https://www.wireshark.org/#download>

Wireshark for Linux**Installing Wireshark on Linux can be a little different depending on the Linux distribution.****From a terminal prompt, run these commands:**

```
sudo apt-get install wireshark
sudo dpkg-reconfigure wireshark-common
sudo adduser $USER wireshark
```

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Those commands download the package, update the package, and add user privileges to run Wireshark.

Capturing Data Packets on Wireshark

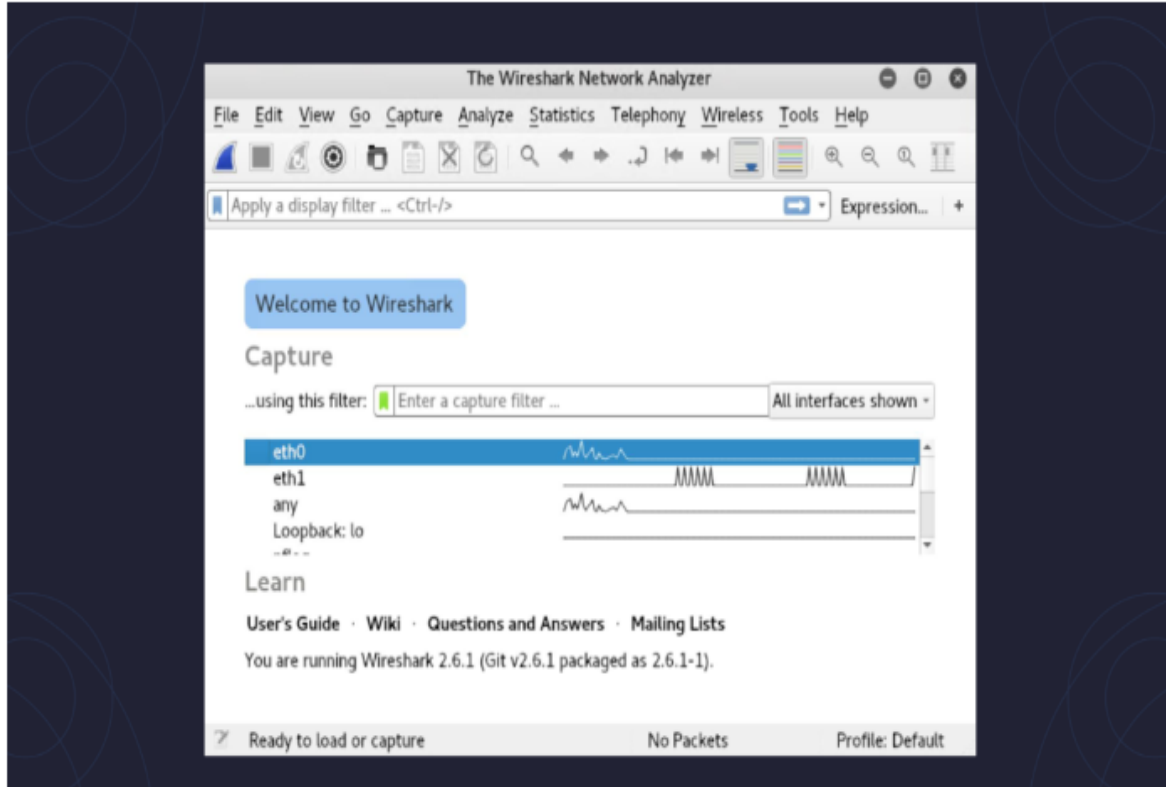


Figure - 7.1

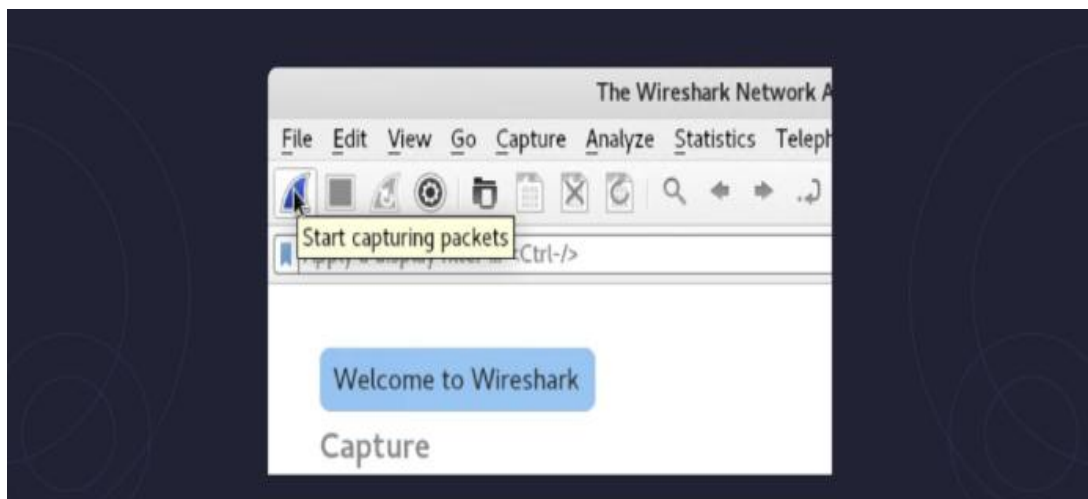


Figure -7.2

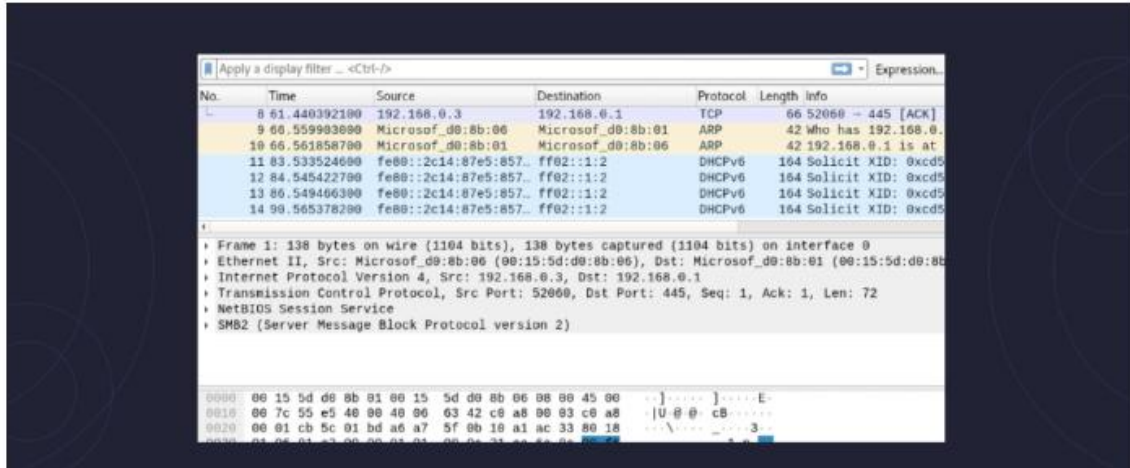


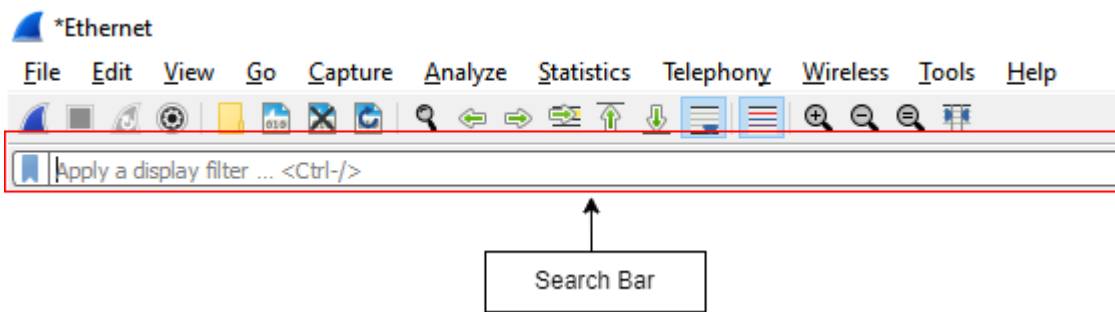
Figure - 7.3

Exercise

1. Turn on Wireshark.
2. Then select your respective network adapter (eg. Ethernet) and start (🔍) capturing the data.
3. Now, go to your browser.
4. Type **172.16.12.25** in your search engine.
5. After 2 seconds go to Wireshark and stop (⏏) capturing data.
6. Go to the Wireshark File menu and click on Save.
7. Save your file as pr-7_xyz_sample1.
8. Repeat steps 1,2,3.
9. Type **172.16.0.1:8090** in your search engine.
10. After 2 seconds go to Wireshark and stop (⏏) capturing data.
11. Go to the Wireshark File menu and click on Save.
12. Save your file as pr-7_xyz_sample2.

(where xyz is last three digits of your enrolment number.)

Based on the above steps, Insert screenshots of sample 1:



Use Different Filters and insert screenshots for sample 1.

The image shows the Wireshark interface with the display filter set to "http". The packet list shows several HTTP continuation packets.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	199.232.214.172	192.168.137.103	HTTP	2954	Continuation
3	0.003029	199.232.214.172	192.168.137.103	HTTP	2954	Continuation
6	0.022999	199.232.214.172	192.168.137.103	HTTP	1510	Continuation
7	0.022999	199.232.214.172	192.168.137.103	HTTP	1510	Continuation
8	0.022999	199.232.210.172	192.168.137.103	HTTP	1510	Continuation

The image shows the Wireshark interface with the display filter set to "ip.dst==172.16.12.38". The packet list shows several TCP and HTTP packets.

No.	Time	Source	Destination	Protocol	Length	Info
659	2.456498	192.168.137.103	172.16.12.38	TCP	66	51188 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
660	2.456744	192.168.137.103	172.16.12.38	TCP	66	51189 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
663	2.462940	192.168.137.103	172.16.12.38	TCP	54	51189 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
664	2.462978	192.168.137.103	172.16.12.38	TCP	54	51188 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
665	2.464918	192.168.137.103	172.16.12.38	HTTP	520	GET /NLP1.txt HTTP/1.1
669	2.475340	192.168.137.103	172.16.12.38	TCP	54	51189 → 80 [ACK] Seq=467 Ack=2674 Win=65280 Len=0

The image shows the Wireshark interface with the display filter set to "ip.src==192.168.137.103". The packet list shows several TCP packets.

No.	Time	Source	Destination	Protocol	Length	Info
634	1.831332	192.168.137.103	199.232.210.172	TCP	66	51048 → 80 [ACK] Seq=445 Ack=105764 Win=137 Len=0 TSval=1119806 TSer=1441294884
635	1.840807	192.168.137.103	199.232.210.172	TCP	66	[TCP ZeroWindow] 51123 → 80 [ACK] Seq=445 Ack=271828 Win=0 Len=0 TSval=1119815 TSer=2814442163
637	1.844901	192.168.137.103	199.232.210.172	TCP	66	51048 → 80 [ACK] Seq=445 Ack=108652 Win=126 Len=0 TSval=1119819 TSer=1441294894
639	1.864666	192.168.137.103	199.232.210.172	TCP	66	51048 → 80 [ACK] Seq=445 Ack=111540 Win=115 Len=0 TSval=1119839 TSer=1441294904
641	1.869805	192.168.137.103	199.232.210.172	TCP	66	51048 → 80 [ACK] Seq=445 Ack=114428 Win=104 Len=0 TSval=1119844 TSer=1441294914

The image shows the Wireshark interface with the display filter set to "arp". The packet list is currently empty.

No.	Time	Source	Destination	Protocol	Length	Info
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File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help					
tcp					
No.	Time	Source	Destination	Protocol	Length Info
1	0.000000	199.232.214.172	192.168.137.103	HTTP	2954 Continuation
2	0.000061	192.168.137.103	199.232.214.172	TCP	66 51044 → 80 [ACK] Seq=1 Ack=2889 Win=437 Len=0 TSval=1117974 TSecr=367618634
3	0.003029	199.232.214.172	192.168.137.103	HTTP	2954 Continuation
4	0.003097	192.168.137.103	199.232.214.172	TCP	66 51044 → 80 [ACK] Seq=1 Ack=5777 Win=437 Len=0 TSval=1117977 TSecr=367618639
5	0.017220	192.168.137.103	49.44.206.90	TCP	54 51186 → 443 [ACK] Seq=1 Ack=1 Win=253 Len=0

Observe the capture data and insert screenshots for sample 2.

Use Different Filters and insert screenshots for sample 2.

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help					
ip.src==192.168.137.103					
No.	Time	Source	Destination	Protocol	Length Info
1	0.000000	192.168.137.103	20.198.119.84	TCP	55 49413 → 443 [ACK] Seq=1 Ack=1 Win=251 Len=1
4	3.006182	192.168.137.103	20.198.119.84	TCP	55 [TCP Keep-Alive] 49413 → 443 [ACK] Seq=1 Ack=1 Win=251 Len=1
5	5.008456	192.168.137.103	142.250.183.131	TCP	55 50939 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
6	6.009914	192.168.137.103	20.198.119.84	TCP	55 [TCP Keep-Alive] 49413 → 443 [ACK] Seq=1 Ack=1 Win=251 Len=1
7	6.305251	192.168.137.103	162.19.234.38	TCP	55 51247 → 443 [ACK] Seq=1 Ack=1 Win=254 Len=1

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help					
ip.dst==172.16.0.1					
No.	Time	Source	Destination	Protocol	Length Info
623	12.107628	192.168.137.103	172.16.0.1	TCP	66 51336 → 8090 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
624	12.107866	192.168.137.103	172.16.0.1	TCP	66 51337 → 8090 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
627	12.112261	192.168.137.103	172.16.0.1	TCP	54 51336 → 8090 [ACK] Seq=1 Ack=1 Win=65280 Len=0
628	12.112328	192.168.137.103	172.16.0.1	TCP	54 51337 → 8090 [ACK] Seq=1 Ack=1 Win=65280 Len=0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help					
arp					
No.	Time	Source	Destination	Protocol	Length Info

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help					
tcp					
No.	Time	Source	Destination	Protocol	Length Info
1	0.000000	192.168.137.103	20.198.119.84	TCP	55 49413 → 443 [ACK] Seq=1 Ack=1 Win=251 Len=1
4	3.006182	192.168.137.103	20.198.119.84	TCP	55 [TCP Keep-Alive] 49413 → 443 [ACK] Seq=1 Ack=1 Win=251 Len=1
5	5.008456	192.168.137.103	142.250.183.131	TCP	55 50939 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
6	6.009914	192.168.137.103	20.198.119.84	TCP	55 [TCP Keep-Alive] 49413 → 443 [ACK] Seq=1 Ack=1 Win=251 Len=1
7	6.305251	192.168.137.103	162.19.234.38	TCP	55 51247 → 443 [ACK] Seq=1 Ack=1 Win=254 Len=1
8	6.363721	162.19.234.38	192.168.137.103	TCP	54 443 → 51247 [ACK] Seq=0 Ack=2 Win=501 Len=0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help					
http					
No.	Time	Source	Destination	Protocol	Length Info
629	12.112480	192.168.137.103	172.16.0.1	HTTP	499 GET /httpclient.html HTTP/1.1
636	12.121253	172.16.0.1	192.168.137.103	HTTP	631 HTTP/1.1 200 OK (text/html)
667	16.125624	192.168.137.103	172.16.0.1	HTTP	553 POST /login.xml HTTP/1.1 (application/x-www-form-urlencoded)
678	16.220211	172.16.0.1	192.168.137.103	HTTP/X...	485 HTTP/1.1 200 OK
682	16.226144	192.168.137.103	172.16.0.1	HTTP	455 GET /images/check-circle.png HTTP/1.1
686	16.230332	172.16.0.1	192.168.137.103	HTTP	361 HTTP/1.1 200 OK (PNG)